دانشگاه صنعتی خواجه نصیر الدین طوسی

K. N. TOOSI UNIVERSITY OF TECHNOLOGY

۱۳۰۷

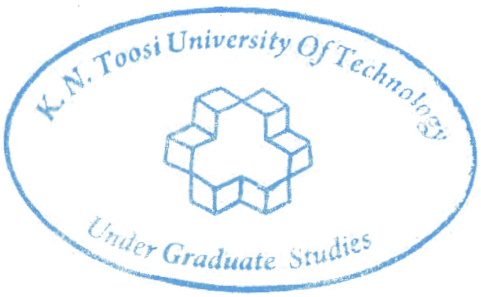
K. N. Toosi University of Technology

Faculty of Mathematics

Course Description Handbook

Undergraduate Program

Mathematics and Applications & Computer Science Minor



Contents

Computer Science Minor (Joint Program with the Faculty of Mathematics) **1**

1st Semester 2018-2019 session (Fall) **2**

Calculus I . . . 2

Fundamental of Computer and Programming . . . 3

Foundations of Mathematics . . . 4

Physics I . . . 6

2nd Semester 2018-2019 session (Winter) **7**

Calculus II . . . 7

Advanced Programming . . . 8

Foundations of Probability . . . 9

Elements of Set Theory . . . 11

3rd Semester 2019-2020 session (Fall) **12**

Calculus III . . . 12

Differential Equations . . . 13

Foundations of Matrices and Linear Algebra . . . 15

Foundations of Combinatorics . . . 17

Probability I . . . 19

4th Semester 2019-2020 session (Winter) **20**

Data Structures and Algorithms . . . 20

Design and Analysis of Algorithms . . . 21

Elementary Mathematical Analysis . . . 23

Foundations of Numerical Analysis . . . 24

5th Semester 2020-2021 session (Fall) **25**

Foundations of Mathematical Analysis . . . 25

Elementary Number Theory . . . 26

Foundations of Algebra . . . 27

Linear Optimization . . . 28

Statistical Methods . . . 29

Coding Theory . . . 30

6th Semester 2020-2021 session (Winter) **32**

Foundation of Geometry 32

Graph Theory and Its Applications 33

Stochastic Processes 34

Theoretical Foundations of Computability 36

Elementary Cryptography 38

Specialized English for Mathematics 40

7th Semester 2021-2022 session (Fall) **41**

Algebra 41

Partial Differential Equations 43

Numerical Linear Algebra 44

Logic Circuits 45

8th Semester 2021-2022 session (Winter) **46**

Elementary Algebraic Geometry 46

Complex Functions 48

General Topology 50

Mathematical Software 52

Computer Science Minor (Joint Program with the Faculty of Mathematics)

A. Completion of at least **six courses** from the following table is required.

* Courses in parentheses are equivalent to the courses listed before the parentheses.

Row	Course Title	Credits	Prerequisite
1	Fundamental of Computer and Programming	3	–
2	Discrete Mathematics (*Foundations of Combinatorics*)*	3	Co-requisite: Fundamentals of Computer and Programming
3	Data Structures and Algorithms	3	Advanced Programming
4	Formal Languages and Automata Theory (*Theoretical Foundations of Computability*)*	3	Co-requisite: Data Structures and Algorithms
5	Design and Analysis of Algorithms	3	Data Structures and Algorithms
6	Engineering Probability and Statistics	3	Calculus I
7	Graph Theory and Its Applications	3	Discrete Mathematics (*Foundations of Combinatorics*)*
8	Operating Systems	3	Computer Architecture
9	Foundations of Matrices & Linear Algebra	3	–
10	Linear Optimization	3	Foundations of Matrices & Linear Algebra
11	Numerical Computation (*Foundations of Numerical Analysis*)*	3	Calculus I

B. In addition to the six compulsory courses listed above, up to **two additional courses** may be selected from the following table.

Row	Course Title	Credits	Prerequisite
1	Logic Circuits	3	Co-requisite: Discrete Mathematics (*Foundations of Combinatorics*)*
2	Computer Architecture	3	Logic Circuits
3	Artificial Intelligence and Expert Systems	3	Design and Analysis of Algorithms
4	Advanced Algorithms	3	Design and Analysis of Algorithms
5	Foundations of Game Theory	3	Design and Analysis of Algorithms
6	Theory of Computation	3	Formal Languages and Automata Theory (*Theoretical Foundations of Computability*)*
7	Coding Theory	3	Discrete Mathematics (*Foundations of Combinatorics*)*
8	Elementary Cryptography	3	Discrete Mathematics (*Foundations of Combinatorics*)*
9	Combinatorial Optimization	3	Discrete Mathematics (*Foundations of Combinatorics*)* and Linear Optimization
10	Numerical Linear Algebra	3	Numerical Computation (*Foundations of Numerical Analysis*)*

1st Semester 2018-2019 session (Fall)

Calculus I

Course title	**Calculus I**	
Prerequisite courses	-	**Course Nature**
		Theoretical ■ Practical □
		Computer Science Minor □
Number of credits	4	
Number of hours	68	
Course classification	Basic Course	

General Purpose

-

Course Description

- Introduction to real and complex numbers;
- Introduction to limit, continuity, differentiation and integration of functions, and their applications;
- Differentiation: Differentiation Rules; Inverse Function and its Derivative; Derivatives of Trigonometric Functions and those of their Inverse Functions; Derivatives of Composite Functions (the Chain Rule);
- Applications of the Derivative: Fermat's Theorem on stationary points; Rolle's Theorem; Mean Value Theorem; Cauchy's Mean-value Theorem; Derivative Tests; Concavity and Points of Inflection; Differentials; Taylor Series Expansion; Physical and Geometric Applications of the Derivative;
- The Transcendental Functions and their Derivatives: The Exponential Function; Natural Logarithm Function; Hyperbolic Functions;
- Integration: The Definite Integral; Mean Value Theorem for Integrals; The Fundamental Theorems of Calculus; Antiderivative/Primitive Functions; Basic Integration Rules;
- Applications of the Integral: Area between Curves; Arc Length; Surface Area; Average Value of a Function; Work; Center of Mass;
- Sequences and Series: Number Sequences; Convergence Theorems (namely, the Ratio and Root Tests); Taylor's Theorems (With/Without Remainder).

References

- Apostol, T. M., *Calculus, vol. 1: One-Variable Calculus, with an Introduction to Linear Algebra, 2nd Ed.*, John Wiley & Sons.
- Thomas, G. B., and Finney, R. L., *Calculus & Analytic Geometry, part 1, 3rd Ed.*, Addison-Wesley Pub. Co.
- Stewart, J., *Calculus, 8th Ed.*, Cengage Learning.

Fundamental of Computer and Programming

Course title	**Fundamental of Computer and Programming**	
Prerequisite courses	-	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Basic Course	

General Purpose

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Course Description

- History of Computer, an introduction on computer structure, general introduction of hardware components of a computer as a computational model, relationship between different components, expression of the simplest basic operations performed by this computational model,
- Introduction of simple algorithms based on this basic operation, review on algorithms for simple problems such as: addition of several numbers - average - search, etc.,
- Introduction of a high-level programming language like Pascal, Java, Python, C or C++ to run the proposed algorithms, familiarity with program structure , introduction of variables and constants, Introducing data types
- Conditional and control structures, types of loops, functions and parameters, working with arrays and files, the concept of runtime and memory complexity, the concept of recursive algorithms, search and sorting algorithms.
- Doing a practical project regarding the field of study

References

- T. H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein, *Introduction to algorithms*, The MIT Press, 2001.
- John V. Guttag, *Introduction to Computation and Programming Using Python, Second Edition: With Application to Computational Modeling and Understanding Data*, The MIT Press, 2016.
- P. J. Deitel and H. M. Deitel, , *C++ How to Program*, Prentice Hall.

Foundations of Mathematics

Course title	**Foundations of Mathematics**	
Prerequisite courses	-	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

Introduction to logic and propositions; truth tables; and elementary methods of proof (including statements of the form $P \to Q$ and related forms). Introduction to sets and operations on sets; relations and functions; and the sets of integers, rational numbers, and real numbers, and prime numbers.

Course Description

- A brief introduction to logic and methods of proof (explicit use of these methods should be practiced in later parts of the course).
- Introduction to sets and their representations; subsets; collections of sets (in particular the power set $P(X)$).
- Union and intersection (especially infinite); set difference and complement; their properties and their relations to each other and to $\subseteq$.
- Cartesian product; coproduct / disjoint union (especially infinite). Some further properties of sets.
- Relations: definition, composition, and inverse relations.
- Order relations; diagrams (Hasse diagrams); ordered sets; greatest and least elements; maximal and minimal elements; upper and lower bounds; introduction to lattices and Boolean algebra with abstract and concrete examples .
- Connections among logic, sets, and Boolean algebra with computer applications (switches and circuits) .
- Equivalence relations, partitions, and their connections .
- Functions (mappings): definition; well-definedness; multivariable functions; n-ary operations; sequences .
- Direct and inverse images and their effect on union, intersection, etc. .
- Composition of functions; universal properties of projection maps from products and inclusion maps into coproducts .
- Introducing the projections (factors) of a product, coproduct (disjoint union), and powers of a set .
- Injective, surjective, and bijective functions; composition rules; cancellation laws; inverses and existence questions .
- Isomorphism (invertibility) and its equivalence with bijectivity; proof of these as practice:

$$\mathbb{N} \neq \mathbb{R}, \quad \mathbb{Z} \approx \mathbb{Q}, \quad P(A) \approx 2^A, \quad A^2 \approx A \times A, \quad A \approx A \times \{1\},$$

$$A^{B \times C} \approx (A^B)^C, \quad [a, b] \approx \mathbb{R} \approx 2^{\mathbb{N}} \approx P(\mathbb{N}), \quad A \not\approx P(A).$$

Further topics of this type

- kernel of functions; the fundamental theorem of functions (isomorphism of the quotient by the kernel with the image); decomposition theorem for functions; several examples .
- Schröder–Bernstein theorem (for proving isomorphism/equinumerosity of sets) (outline of proof); several examples .
- Finite and infinite sets; countable and uncountable sets; related theorems and examples .
- Cardinal numbers; ordering and operations on them .
- Statement of the Axiom of Choice and some of its equivalents (especially Zorn's Lemma); examples of its applications .

References

- Shwu-Yeng T. Lin, *Set Theory With Applications*, Mancorp Pub, 1985.
- D. C. Goldrei, *Classic Set Theory*, Chapman & Hall/CRC Press, 1996.
- Paul R. Halmos, *Native Set Theory*, Springer-Verlag, VII, 1974.

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Physics I

Course title	**Physics I**	
Prerequisite courses	-	**Course Nature**
		Theoretical ■ Practical □
		Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Basic Course	

General Purpose

The primary goal of the introductory Physics I course is to provide students with the tools needed to study scientific topics. Students should learn the fundamental concepts, be able to answer scientific questions, and solve physics problems. In Physics I, the first part of the course introduces basic concepts in classical mechanics and thermodynamics.

Course Description

- Measurement, quantities and units, precision, dimensional analysis
- Motion in one dimension; velocity, acceleration
- Motion in plane; velocity and acceleration in two dimensions
- Force and Newton's laws
- Dynamics of circular motion; oscillation and Hook's law
- Work and energy; conservation of energy, kinetic and potential energy
- System of particles; center of mass, momentum, conservation of momentum and collision
- Rotational kinematics and dynamics
- Fluid mechanics; hydrostatic pressure, conservation laws in fluid motion; viscosity
- Thermodynamics and heat
- Waves

References

- *Fundamentals of Physics Extended, 10th ed.*, D. Halliday, R. Resnick, J. Walker, (August 5, 2013), Wiley.
- *Physics, Principles with Applications, 7th ed.*, D.G. Giancoli, Prentice Hall, (2014).
- *University Physics with Modern Physics*, Technology Update, 13th ed., H.D. Young and R.A. Freedman, (2013).

2nd Semester 2018-2019 session (Winter)

Calculus II

Course title	**Calculus II**	
Prerequisite courses	Calculus I	**Course Nature**
		Theoretical ■ Practical □
		Computer Science Minor □
Number of credits	4	
Number of hours	68	
Course classification	Basic Course	

General Purpose

This is a continuation of Calculus I. It considers more advanced topics in calculus such as multiple integrals, and vector analysis.

Course Description

- Parametric equations; coordinates of three-dimensional space; vectors in three-dimensional space; scalar product; 3*3 matrices; linear transformations and their associated matrices; system of linear equations in three variables; row operations; inverse of a matrix; solution of linear systems; linear independence; Basis; determinants; eigenvalues and eigenvectors; vector product in three dimensional space; equations of lines and planes;
- Quadratic surfaces; vector functions and their derivative; velocity and acceleration; curvature and normal vectors; multi-variable functions; directional derivatives and partial derivatives; tangent plane and normal line; gradient; chain rules for partial derivatives; exact differential;
- Double and triple integrals, and their applications to geometrical and physical problems; changing the order of integrations (without a complete proof); cylindrical and spherical coordinates; vector fields; line integrals;
- Surface integrals; divergence; curl; Lapalacian; potential; green's Theorem, Divergence theorem, and Stokes' theorem.

References

دانشگاه صنعتی خواجه نصیرالدین طوسی

- *Calculus, Vol. 2: Multi-Variable Calculus and Linear Algebra with Applications to Differential Equations and Probability; 2nd Ed*, Apostol, T. M., John Wiley & Sons.
- *Calculus & Analytic Geometry, part 2, 3rd Ed.*, Thomas, G. B., and Finney, R. L, Addison-Wesley Pub. Co.

Advanced Programming

Course title	**Advanced Programming**	
Prerequisite courses	Fundamental of Computer and Programming	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

The main purpose of this course is to learn the principles of advanced programming. Given that the prerequisite for this course is the fundamentals of computer science and programming, it is necessary for students who choose this course to have basic knowledge in the field of programming and have acquired the necessary skills in this field to the extent of the basic concepts required for object-oriented programming.

Course Description

Introduction to programming, familiarity with programming languages and programming language selection criteria, familiarity with a programming language such as Java or C ++, object-oriented programming, object and class definition, inheritance, access levels, encapsulation, overriding and overloading methods, statistic methods and variables, IO structures, graphical user interface (GUI) design, error handling (Exception Handling), working with files, working with data structures (Array, ArrayList, HashMap, HashSet, Vector, ...), project management and teamwork skills, proposal writing and relevant issues, documenting programs, reviewing algorithms and solving some sample problems, familiarity with the complexities of algorithms and time complexity order, familiarity with search and sorting algorithms and how to analyze them theoretically.

References

- T. H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein, *Introduction to algorithms*, The MIT Press, 2001.
- John V. Guttag, *Introduction to Computation and Programming Using Python, Second Edition: With Application to Computational Modeling and Understanding Data*, The MIT Press, 2016.
- P. J. Deitel and H. M. Deitel, , *C++ How to Program*, Prentice Hall.

Foundations of Probability

Course title	**Foundations of Probability**	
Prerequisite courses	Calculus I	**Course Nature** Theoretical ■ Practical ☐ Computer Science Minor ☐
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

This course aims to introduce the fundamental concepts and tools of descriptive statistics and elementary probability. Students learn how to organize and represent data, compute and interpret measures of central tendency and dispersion, and understand basic probability models and key concepts such as conditional probability, Bayes' formula, and independence.

Course Description

- **Basics of descriptive statistics:** : definition of statistics; applications in other sciences; relation to and role in data science; statistical population and sample; types of variables (qualitative/quantitative; discrete/continuous); measurement scales (nominal and ordinal); types of data (univariate, multivariate, time-series).
- **Data collection and data sources**: data collection methods; recorded data sources; introduction to data (including large-scale data) and their sources.
- **Statistical tables**: data grouping; definition of frequency, cumulative frequency, relative frequency, cumulative relative frequency; class mark / class midpoint (class representative); constructing frequency distribution tables (including for one quantitative variable).
- **Statistical charts/plots**: stem-and-leaf plot; histogram; frequency polygon and cumulative frequency polygon; frequency curves and cumulative-frequency curves; empirical distribution and normal distribution; bar charts; pie charts; time-series plots; scatter plots for bivariate data and displaying correlation; box plot.
- **Measures of central tendency**: arithmetic mean, geometric mean, weighted mean, harmonic mean, root-mean-square (RMS); median; mode; quantiles; quartiles, deciles, percentiles.
- **Measures of dispersion**: range; interquartile range; deviation from the mean; mean absolute deviation (from the mean and from the median); variance; standard deviation; coefficient of variation.
- **Shape measures for a univariate frequency distribution**: symmetry/asymmetry (skewness) and kurtosis (peakedness/flatness).
- **Data summarization and presentation**: summarizing data using measures of central tendency and dispersion; preparing one-way and two-way tables; presenting and interpreting results; drawing histograms, frequency plots, box plots, and other standard statistical charts.
- **Introduction to statistical software packages**: practical work with software for descriptive analysis and for producing the above statistical tables and plots.
- **Basic counting methods**: counting rules; factorial; ordered samples and permutations; unordered samples and combinations; permutations with repeated elements.
- **Probability**: random experiments (simple and compound) and related concepts; sample space

and events (finite, countably infinite, and uncountable); operations on events; different definitions/interpretations of probability; probability function; Kolmogorov axioms; uniform probability space (classical probability model); continuity of the probability function; conditional probability; compound experiments; applications of conditional probability and Bayes' formula; independence.

References

- Robert V. Hogg and Elliot A. Tanis, *Probability and Statistical Inference*, Persian translation by Norooz Irzadoustdar and Hamid Pezeshk, 1381 (Iranian calendar), University of Tehran Press.
- Sheldon Ross, *A First Course in Probability*, Persian translation by Ahmad Parsian and Ali Hamdani, 8th ed., 10th printing, 1389 (Iranian calendar), Sheikh Bahaei Publications.
- G. R. Grimmett and D. J. A. Welsh, *Probability: An Introduction*, Clarendon Press, 1986.

دانشگاه صنعتی خواجه نصیرالدین طوسی

Elements of Set Theory

Course title	**Elements of Set Theory**	
Prerequisite courses	Foundations of Mathematics	**Course Nature** Theoretical ■ Practical ☐ Computer Science Minor ☐
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

To familiarize students with mathematical logic and the introductory concepts of set theory, and to develop the ability to formally describe and verify the correctness of mathematical systems (or computer systems) using the formal tools presented in this course.

Course Description

- **Introduction to Logic**
  - Propositional logic; the language of propositional logic; natural deduction rules; semantics; the soundness and completeness theorems.
  - Normal forms and SAT algorithms.
  - Predicate logic; the language of predicate logic; natural deduction rules; expressibility/definability of properties in a language.
  - Introduction to the *Prolog* language .
- **Introduction to Set Theory**
  - Review of set operations: union and intersection; complement; taking complements; definition of function and relation; axioms/principles of set theory; Russell's paradox.
- **Set Theory as a Foundation**
  - Construction of the natural numbers; construction of the rational numbers; construction of the real numbers.
- **Infinite Sets**
  - Cardinal numbers; ordinal numbers; well-ordering.

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References

- lin, Shwu-Yeng T., *Set Theory With Applications*, Mancorp Pub, 1985.
- J. M. Henle, *An Outline of Set Theory*, Springer-Verlag, 1986.
- M. Huth and M. Ryan, *Logic in Computer Science: Modeling and Reasoning about Systems*, Cambridge University Press, 2004.

3rd Semester 2019-2020 session (Fall)

Calculus III

Course title	**Calculus III**	
Prerequisite courses	Calculus II	**Course Nature**
		Theoretical ■ Practical □
		Computer Science Minor □
Number of credits	4	
Number of hours	68	
Course classification	Basic Course	

General Purpose

This course is about multivariate calculus and vector analysis with a geometric approach.

Course Description

- Derivatives of functions of several variables.
- Taylor polynomials in several variables.
- Mean value theorem in several variables.
- Inverse function theorem.
- Implicit function theorem.
- Critical points and Morse's lemma.
- Multiple integrals.
- Change of variables in multiple integrals.
- Fubini's theorem.
- Line and surface integrals.
- Differential forms.
- Green's theorem and the divergence theorem.
- Stokes' theorem.
- Closed and exact forms.
- Poincaré lemma.

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References

- James J. Callahan, *Advanced Calculus: A Geometric View*, Springer, 2010.
- Peter D. Lax and Maria Shea, *Multivariable Calculus with Applications*, Springer, 2017.
- Ib H. Madsen and Jørgen Tornehave, *From Calculus to Cohomology: De Rham Cohomology and Characteristic Classes*, Cambridge University Press, 1997.
- James Stewart, Daniel K. Clegg, and Saleem Watson, *Multivariable Calculus*, Cengage Learning, 2020.

Differential Equations

Course title	**Differential Equations**	
Prerequisite courses	Calculus I	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

The aim of this course is to introduce the fundamental concepts of ordinary differential equations, emphasize existence and uniqueness principles, and develop students' ability to solve first- and second-order differential equations, analyze series solutions, and apply integral transforms (especially the Laplace transform) to the solution of differential equations and linear systems.

Course Description

- **Introduction to Differential Equations:** General notes on solutions of differential equations. Classification of differential equations. Existence and Uniqueness Theorem.
- **First-Order Differential Equations:** Separable equations, Homogeneous equations, Equations reducible to homogeneous. Exact equations and Integrating Factors (functions of x, y, $x^{\alpha}y^{\beta}$, and $\mu(z)$ where z is known). First-order linear equations. Important non-linear equations (Bernoulli, Lagrange, etc.). Family of curves and Orthogonal Trajectories. Modeling with first-order equations.
- **Second-Order Differential Equations:** Reduction of order. Preliminary concepts of linear equations. General solution of homogeneous and non-homogeneous linear equations. Using a known solution to find another solution. Linear homogeneous equations with constant coefficients (second-order and higher). Non-homogeneous linear equations. Operator methods for equations with non-constant coefficients (Cauchy-Euler equations, etc.). Introductory theory of boundary value problems (Eigenvalues and Eigenfunctions, etc.).
- **Power Series Solutions and Special Functions:** Review of power series. Solutions around ordinary points. Legendre's equation and Legendre polynomials. Properties of Legendre polynomials. Solutions around singular points (Frobenius method). Bessel's equation, Gamma function, and properties of Bessel functions.
- **The Laplace Transform and Its Applications:** Introduction (Notes on Laplace Theory). Existence Theorem. Laplace Transform of derivatives and integrals. Translation (Shifting) Theorems. Introduction to Step Functions and the Dirac Delta Function. Applications of Laplace Transforms in differential equations. Derivatives and integrals of Laplace Transforms. Introduction to Convolution. Introduction to integral equations. Solving systems of linear equations using Laplace Transforms.
- **Systems of differential equations:** introduction to linear systems; solving homogeneous and nonhomogeneous linear systems with constant coefficients; methods of eigenvalues and eigenfunctions.

References

- George F. Simmons, *Differential Equations with Applications and Historical Notes* (3rd Edition),

Chapman and Hall/CRC, 2016.

- Masoud Nikookar, *Ordinary Differential Equations*, Azadeh Publication, 2013.

دانشگاه صنعتی خواجه نصیرالدین طوسی

Foundations of Matrices and Linear Algebra

Course title	**Foundations of Matrices and Linear Algebra**	
Prerequisite courses	Foundations of Mathematics	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

Familiarity with matrices and vector spaces and their elementary applications in solving systems of linear equations, and preparing students to utilize these tools in other pure and applied courses. This fundamental course is mandatory for all majors and concentrations in the mathematical sciences. Its syllabus covers three main topics: matrices, vector spaces, and their connections and applications. Therefore, both the pure and applied aspects of these concepts must be considered, as neglecting either one reduces the effectiveness of this course.

Course Description

- **Matrices and Systems of Equations**: Introduction to algebraic systems of matrices ; definition and operations, transpose, and inverse; types of matrices (zero, identity, diagonal, triangular, symmetric, and skew-symmetric); matrix rank and methods for determination; row and column equivalence and methods for determination; Gaussian elimination and elementary row operations; elementary matrices and their properties; calculating matrix inverses ; discussion on solving systems of linear equations ; determinants: calculation and properties ; Cayley–Hamilton theorem; rank of matrix determinants.
- **Vector Spaces** : Vector spaces and examples; subspaces, products, quotient spaces, and direct sums ; linear independence, basis, and dimension ; row space and column space of a matrix.
- **Linear Transformations and Matrices** : Definition, examples, and elementary properties ; kernel, image, and the fundamental theorem (isomorphism of quotient domain to image) ; space of linear transformations and functionals ; matrix representation of linear transformations ; change of basis ; rank of linear transformations ; eigenvectors, eigenvalues, and characteristic polynomials ; diagonalizable matrices and triangularization ; inner product spaces: definitions, orthogonality, and orthonormal sets ; Gram–Schmidt orthonormalization process and QR decomposition; applications of linear algebra in other sciences (optional).

References

- *Linear Algebra*, author: Michael Spivak; translator: Ali Akbar Mohammadi Hasanabadi.
- *Linear Algebra*, authors: Kenneth Hoffman and Ray Kunze; translator: Jamshid Farshidi.
- *Linear Algebra*, authors: Gene H. Golub and Charles F. Van Loan; translators: Mohammad Reza Darvishi and Dr. Negar Shatti-Karmazadeh.
- *Linear Algebra (Textbook) with Applications to Differential Equations*, author: Daniel B. Murray; translator: Dr. Malek Mansour Sharif.
- *Linear Algebra*, author: Serge Lang; translator: Mohammad Hossein Tarhkhani; publisher: Kourosh.
- *Linear Algebra and Its Applications*, author: Gilbert Strang; translator: Mojtaba Rezaei Torabi

and Ebrahim Yazdani Bozorgi-Nia.

- *Linear Algebra with Applications to Differential Equations*, authors: G. C. Jain and K. K. Kaushal; translators: Professor Ali Akbar Alizadeh and Professor Mohammad Hossein Mirzazadeh.

دانشگاه صنعتی خواجه نصیرالدین طوسی

Foundations of Combinatorics

Course title	**Foundations of Combinatorics**	
Prerequisite courses	Calculus I	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

The primary objective of this course is to familiarize the student with the main and foundational concepts in combinatorics and discrete mathematics. This is done in such a way that the student, while becoming acquainted with these concepts, also becomes familiar with the applications and core motivations that exist within the science of combinatorics, and encounters some of its main problems as well.

In this regard, and considering the content and nature of this course, other objectives can also be pursued, which include:

a) Practicing the presentation of precise mathematical reasoning and their various types (e.g., mathematical induction, proof by contradiction, etc.)

b) Familiarization with combinatorial reasoning (e.g., arguments based on counting, the extremal principle, etc.)

c) Familiarization with various discrete structures (e.g., finite sets, finite relations, matrices, graphs, Latin squares, etc.)

d) Practicing the presentation of reasoning based on algorithmic thinking and familiarization with various algorithms in the field of combinatorics.

Course Description

- **A quick review of set concepts**, functions, algorithms, and propositional logic and Boolean algebra (coordinated with the course "Foundations of Mathematics").
- **Counting including basic concepts:**, the Pigeonhole Principle, permutations and combinations, binomial coefficients, Inclusion–Exclusion Principle, recurrence relations, and generating functions.
- **Relations and their types**: relations and their representation, equivalence relations and partitions, partial and total order relations, closure of a relation with respect to various properties (this section is presented in coordination with the "Foundations of Mathematical Sciences" course to avoid repetition).
- **Matrices**: matrices from a combinatorial perspective, especially certain important properties of zero-one matrices (preparation for the sections on Latin squares and graphs), familiarity with Hadamard matrices and some results in this area (at the instructor's discretion).
- **Graphs and models based on them**: introduction to the concept of a graph with an emphasis on its applications in modeling (with several examples at the instructor's discretion), familiarity with basic concepts of graph theory such as paths, cycles, degrees, degree sequences, types of graphs such as complete graphs, trees, bipartite graphs, Eulerian and Hamiltonian graphs, and directed graphs and tournaments (emphasizing examples and applications), matchings and max-

imum matchings (algorithm design and applications), graph coloring and chromatic polynomials (presenting examples and algorithms).

- **Latin squares, designs, and finite geometries**: introduction to definitions and basic concepts with an emphasis on the connection between these topics (by providing examples) and also emphasizing the connection of these concepts with topics previously presented in graph theory and counting, presenting the concept of Systems of Distinct Representatives (SDR) and also presenting Hall's Theorem and providing examples and applications in Latin squares and several practical applications (at the instructor's discretion).

References

- I. Anderson, *A First Course in Combinatorial Mathematics*, 2nd ed., Oxford Applied Mathematics and Computing Science Series, Clarendon Press, Oxford University Press, New York, 1989.
- M. Erickson, *Pearls of Discrete Mathematics*, Discrete Mathematics and Its Applications, CRC Press, Boca Raton, FL, 2010.
- R. Garnier and J. Taylor, *Discrete Mathematics for New Technology*, 2nd ed., IOP Publishing Ltd., Bristol, 2002.
- R. Garnier and J. Taylor, *Discrete Mathematics*, 3rd ed., CRC Press, Boca Raton, FL, 2010.
- L. Lovász, J. Pelikán, and K. Vesztergombi, *Discrete Mathematics*, Undergraduate Texts in Mathematics, Springer-Verlag, New York, 2003.

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Probability I

Course title	**Probability I**	
Prerequisite courses	Foundations of Probability	**Course Nature**
		Theoretical ■ Practical ☐
		Computer Science Minor ☐
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

Course Description

- **Random Variables:**
  - Definition of random variables, distribution functions, discrete and continuous random variables, and distributions of functions of random variables.
- **Standard Distributions:**
  - Bernoulli, Binomial, Geometric, Hypergeometric, Negative Binomial, and Poisson.
  - Uniform (discrete and continuous), Exponential, Gamma (+ Chi-squared), Normal, Beta, Cauchy, Logistic, Weibull, and Pareto.
- **Expectation and Moments:**
  - Expected value and its properties, expectation of a function of a random variable, and Jensen's inequality.
  - Median and mode of a distribution, variance, and other dispersion measures.
  - Symmetry, skewness, and kurtosis; moments of a distribution.
- **Generating Functions and Inequalities:**
  - Moment generating functions and probability generating functions.
  - Important probability inequalities including Markov and Chebyshev inequalities.
- **Multivariate Distributions:**
  - Multivariate random variables, joint discrete and continuous distributions of two variables, independent random variables, covariance, and correlation coefficient.

References

- Ross, S., *A First Course in Probability*, translated by Ahmad Parsian and Ali Hamadani, 8th Edition, 2010 (1389 AP), Sheikh Baha'i Publications.
- Ghahramani, S., *Fundamentals of Probability*, translated by Shahkar, 1st Edition, 2001 (1380 AP), Ferdowsi University of Mashhad Publications.
- G. R. Grimmett and D. Stirzaker, *Probability and Random Processes*, 3rd Ed., Oxford, 2001.

4th Semester 2019-2020 session (Winter)

Data Structures and Algorithms

Course title	**Data Structures and Algorithms**	
Prerequisite courses	Advanced Programming	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

Course Description

- **Introduction to algorithms and data structures:** definition and importance of data structures in problem solving; analysis of algorithms with emphasis on time and memory complexity.
- Static data structures including variables and arrays and their applications.
- Semi-static data structures including stacks and queues, methods of representing stacks and queues, and their applications.
- Dynamic data structures including linked lists and their applications.
- **Trees and their applications:** trees; binary trees; binary tree traversal methods; binary search trees; threaded binary trees; expression trees; different representations of expressions.
- **Graphs and graph algorithms:** types of graphs; graph representations; graph traversal and search algorithms including Breadth-First Search (BFS) and Depth-First Search (DFS); applications of graph algorithms.
- **Searching algorithms:** linear search and binary search; tree-based searching methods including AVL trees, Red–Black trees, Splay trees, and Trie structures; pattern matching.
- Priority queues; heap-based structures and algorithms.
- Hashing algorithms and hash functions; hashing and its applications.
- **Sorting algorithms:** stable and unstable sorting methods; sorting algorithms including Merge Sort, Quick Sort, Insertion Sort, Bubble Sort, and Radix Sort; internal sorting methods; statistical comparison of sorting algorithms; external sorting.
- Recursive functions and their role in algorithm design.
- **Fundamental algorithm design techniques:** backtracking; greedy algorithms; dynamic programming.

References

- T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 3rd ed., MIT Press, 2009.
- A. V. Aho, J. E. Hopcroft, and J. D. Ullman, *Data Structures and Algorithms*, Addison–Wesley Series in Computer Science and Information Processing, 1983.
- U. Manber, *Introduction to Algorithms: A Creative Approach*, Addison–Wesley Longman Publishing Co., Inc., 1st ed.

Design and Analysis of Algorithms

Course title	**Design and Analysis of Algorithms**	
Prerequisite courses	Foundations of Combinatorics - Theoretical Foundations of Computability	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

Review of basic complexity theory concepts and asymptotic analysis, including the symbols O, o, Ω, ω, and Θ; review of basic data structures; methods for solving recurrence equations.

Lower and upper bounds for time and memory complexity, at the level of algorithms presented in the course, with emphasis on computing and comparing these bounds.

Course Description

- **Fundamental algorithm design paradigms:** recursion and induction; Divide and Conquer (including Quick Sort, Merge Sort, selection of the k-th smallest element, fast multiplication of numbers and matrices); Dynamic Programming (including Longest Common Subsequence, 0/1 Knapsack, optimal binary search trees, and counting binary trees); Greedy algorithms (including fractional knapsack, Huffman coding, scheduling problems).
- **Backtracking and Branch-and-Bound methods:** including applications to problems such as the 0/1 Knapsack problem and the n-Queens problem.
- **Detailed analysis of sorting algorithms:** including Sort, Quick Sort, Heap Sort, and Merge Sort, with emphasis on time complexity analysis and construction of the associated data structures.
- **Numerical and selection algorithms:** including finding Min, Max, Median, and similar problems; multiplication of large numbers and matrices with time complexity estimation.
- Re-emphasis on data structures and their role in algorithm design and analysis, including binary search trees, hash tables, heaps, stacks, and queues, with advanced discussions (at the instructor's discretion) such as B-trees and different types of heaps.
- **Graph algorithms:** Breadth-First Search (BFS) and Depth-First Search (DFS); shortest path algorithms; minimum spanning tree algorithms; connected components; topological sorting; and analysis of their time complexity.
- String matching algorithms.
- **Randomized algorithms:** including Randomized Quick Sort and related techniques, with expected-time analysis.
- **Approximation algorithms:** including approximation algorithms for the Knapsack problem, Vertex Cover, and related problems, with analysis of approximation ratios.

References

- M. H. Alsuwaiyel, *Algorithms: Design Techniques and Analysis*, World Scientific Publishing, 1999.
- T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 3rd ed., MIT Press, 2009.

- D. Kozen, *The Design and Analysis of Algorithms*, Springer-Verlag New York Inc., 1992.
- J. J. McConnell, *Analysis of Algorithms: An Active Learning Approach*, Jones & Bartlett Publishers Inc., 2008.

دانشگاه صنعتی خواجه نصیرالدین طوسی

Elementary Mathematical Analysis

Course title	**Elementary Mathematical Analysis**	
Prerequisite courses	Foundations of Mathematics - Calculus II	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

Mathematical Analysis is a branch of mathematics dealing with limits and convergence. This course aims to acquaint students with metric spaces and related notions such as compactness and connectedness, sequences in metric spaces, and continuity of functions. Introduction to metric spaces. Study of the concept of continuity and its relation to other concepts such as compactness and connectedness.

Course Description

- Real number system; least upper bound (LUB) property; Archimedean property; structure of the real numbers; extended real number system; ordered fields; complex number system.
- Cauchy–Schwarz inequality.
- Metric spaces; open and closed sets; limit points; closure; complete sets.
- Sequences in metric spaces; limits of sequences; Cauchy sequences; subsequences and sets of subsequential limits; lim sup and lim inf.
- Numerical series; series with nonnegative terms; root and ratio tests; alternating series; conditional and absolute convergence; sum, product, and rearrangement of series.
- Compact sets; sequential compactness; Heine–Borel theorem; totally bounded sets; perfect sets.
- Connectedness.
- Bases in metric spaces; separable spaces; Lindelöf theorem.
- Cantor set.
- Sets of the first and second category; Baire category theorem.
- Limits and continuity of functions in metric spaces; theorems on continuity of functions and continuity of inverse functions; intermediate value theorem.
- Uniform continuity; relationship between continuity, compactness, and connectedness.
- Types of discontinuities; discontinuity of monotone functions.
- Lipschitz conditions and fixed point theorems.

References

- Robert G. Bartle and Donald R. Sherbert, *Introduction to Real Analysis*, Wiley, 2011.
- Michael O. Searcoid, *Metric Spaces*, Springer-Verlag, London, 2007.
- W. Rudin, *Principles of Mathematical Analysis*, McGraw–Hill, 1976.

Foundations of Numerical Analysis

Course title	**Foundations of Numerical Analysis**	
Prerequisite courses	Calculus II	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

Numerical analysis contains designing and analyzing numerical algorithms to solve problems such as finding best approximation, integration, derivation, solving linear system of equations, differential equations and so on. These problems appear in many branches of science and engineering. This course is an introduction to some of these algorithms and their convergence analysis.

Course Description

- **Preliminaries of calculus:** Big–O and little–o notation; introduction to mathematical software; order of convergence.
- **Errors and floating point errors:** sources and effects of errors on numerical results; error propagation; methods for reducing and eliminating errors; floating point arithmetic.
- **Numerical methods for solving nonlinear equations:** bisection method; Newton's method; secant method; convergence analysis of iterative methods; contractive mapping theorem; functional (fixed-point) iteration method.
- **Interpolation:** polynomial interpolation; Lagrange interpolation; Newton divided difference interpolation; interpolation errors; Richardson extrapolation; Hermite interpolation; cubic splines and their error analysis.
- **Numerical differentiation and integration:** numerical differentiation using interpolation; Newton–Cotes integration methods including the trapezoidal rule and Simpson's rules; Gaussian quadrature methods; error analysis of numerical integration methods.

References

- D. Kincaid and W. Cheney, *Numerical Analysis*, Cole Publishing Company, 1990.
- E. Babolian, *Basics of Numerical Analysis*, Fatemi Publishing Company, 2013.
- E. Babolian, *Numerical Analysis 1*, Payam Noor Publishing Company, 2011.
- A. Kerayechian, *Numerical Analysis 1*, Ferdowsi University of Mashhad Publishing Company, 2011.

دانشگاه صنعتی خواجه نصیرالدین طوسی

5th Semester 2020-2021 session (Fall)

Foundations of Mathematical Analysis

Course title	**Foundations of Mathematical Analysis**	
Prerequisite courses	Elementary Mathematical Analysis	**Course Nature**
		Theoretical ■ Practical □
		Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

Course Description

- review of the structure of real numbers; sequences and series; limits and continuity; derivatives; the Riemann integral; sequences and series of functions.
- algebraic properties of real numbers; order properties of real numbers; absolute value; completeness property; properties of supremum and infimum; sequences and their limits; limit theorems; monotonic sequences; subsequences and the Bolzano–Weierstrass theorem;
- elementary concepts of metric spaces, including compactness and connectedness; the Cauchy criterion;
- limits of functions; limit theorems; continuous functions; composition of continuous functions; uniform continuity; monotonic functions and inverse functions;
- derivatives of functions; the Mean Value Theorem; approximation methods based on Taylor's theorem;
- Riemann integrability; properties of the Riemann integral; the Fundamental Theorem of Calculus; the integral as a limit; approximate (numerical) integration;
- sequences of functions; pointwise and uniform convergence; interchange of limits;
- convergence of series of functions; absolute and uniform convergence tests;
- power series and Taylor series and their fundamental theorems.

References

- Walter Rudin, *Principles of Mathematical Analysis*, translated by Ali Akbar Alemzadeh.
- Tom M. Apostol, *Mathematical Analysis*, translated by Ali Akbar Alemzadeh, Scientific Publications Institute of Sharif University of Technology, 1359 [1980].
- Robert G. Bartle, *The Elements of Real Analysis*, translated by Dr. Jafar Zafarani, University Publishing Centre, 1366 [1987].

Elementary Number Theory

Course title	**Elementary Number Theory**	
Prerequisite courses	Foundations of Mathematics	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	core Course	

General Purpose

Divisibility, Division Algorithm, GCD (Greatest Common Divisor) and LCM (Least Common Multiple), Fundamental Theorem of Arithmetic, Introduction and study of the congruence ring $\mathbb{Z}_n$ and the multiplicative group $\mathbb{Z}_p^* = \mathbb{Z}_p \setminus \{0\}$. Solving and discussing Diophantine equations, Arithmetic functions. Quadratic Reciprocity Law, applications in Cryptography and Coding.

Course Description

- **Divisibility and Factorization:** divisibility; division algorithm; prime numbers; Fundamental Theorem of Arithmetic; greatest common divisor (GCD); linear representation and the Euclidean algorithm ; solving linear Diophantine equations .
- **Congruences (9 hours):** definition and elementary properties of congruences; residue systems and reduced residue systems; introduction to the ring $\mathbb{Z}_n$ and the multiplicative group C_n of invertible elements of $\mathbb{Z}_n$; the multiplicative group C_p in the language of congruences; linear congruences; systems of linear congruences; Chinese Remainder Theorem ; Fermat's theorem; Euler's theorem; Wilson's theorem and their relationships with the groups C_n and C_p; selected results .
- **Primitive Roots :** definition of multiplicative order modulo n and p (in $\mathbb{Z}_n$ and C_p) and its properties ; primitive roots as generators of the multiplicative group C_p and their existence ; solving and discussion of polynomial congruence equations modulo n, $f(x) \equiv 0$.
- **Arithmetic Functions :** arithmetic functions; multiplicative functions; number and sum of divisors; Möbius function; Euler's totient function $\varphi(n)$; Mersenne primes; perfect numbers; even perfect numbers.
- **Quadratic Residues:** quadratic residues and non-residues and their properties ; Euler's criterion; Gauss's lemma (possibly without proof, with examples) ; Quadratic Reciprocity Law .
- **Other Topics :** continued fractions (finite and infinite) ; representation of integers as the sum of two and four squares ; Pythagorean triples ; Pell's equation ; applications of number theory in cryptography and coding theory .

References

- Kenneth H. Rosen, *Elementary Number Theory*, Pearson Education, London, 2011.
- D. M. Burton, *Elementary Number Theory*, McGraw–Hill Education, 2010.
- Kenneth Ireland and Michael Rosen, *A Classical Introduction to Modern Number Theory*, Springer–Verlag, New York, 1990.
- Ivan Niven, Herbert S. Zuckerman, and Hugh L. Montgomery, *An Introduction to the Theory of Numbers*, John Wiley & Sons, 1991.

Foundations of Algebra

Course title	**Foundations of Algebra**	
Prerequisite courses	Foundations of Mathematics	**Course Nature** Theoretical ■ Practical ☐ Computer Science Minor ☐
Number of credits	3	
Number of hours	51	
Course classification	core Course	

General Purpose

introduction and elementary study of general algebraic structures (ordinary and ordered); quotients and morphisms; isomorphism theorems; detailed study of these concepts with emphasis on groups and rings.

Course Description

- **General Algebraic Structures :** binary operations; n-ary operations (especially 0-, 1-, and 2-ary); binary algebraic structures; isomorphism of binary structures; familiar examples from numbers, functions, matrices, sets, union, intersection, absolute value, logical OR ($\vee$); Cayley tables; subalgebras; products; congruences and quotient algebras; homomorphisms and isomorphisms; varieties and systems of equations (statement of Birkhoff's theorem); ordered algebras and their isomorphisms; groupoids; semigroups; monoids; lattices; free monoids and free groups (words and codes); monoid actions on sets (applications to automata); quasigroups; Latin squares.
- **Groups:** groups and equivalent definitions; subgroups; cyclic groups; order of an element; structure theorems of cyclic groups (finite and infinite); group homomorphisms and isomorphisms; permutation groups; orbits and cycles; symmetric, alternating, and dihedral groups; Cayley's theorem; cosets; Lagrange's theorem; normal subgroups; quotient (factor) groups; center of a group; derived (commutator) subgroup; simple groups; direct products of groups; finitely generated abelian groups; isomorphism theorems of groups.
- **Rings :** examples of rings (including quaternions); subrings; ring homomorphisms; zero divisors; integral domains; characteristic of a ring; embeddings of rings; field of fractions of an integral domain; ideals and generated ideals; factor (quotient) rings; prime ideals; maximal ideals; isomorphism theorems of rings; polynomial rings.

دانشگاه صنعتی خواجه نصیرالدین طوسی

References

- S. Shahriari, *Algebra in Action: A Course in Groups, Rings, and Fields*, American Mathematical Society, 2017.
- D. S. Dummit and R. M. Foote, *Abstract Algebra*, Wiley, 2004.
- T. W. Hungerford, *Abstract Algebra: An Introduction*, Saunders College Publishing, 1996.
- I. N. Herstein, *Abstract Algebra*, Macmillan Company, 1989.
- N. Lauritzen, *Concrete Abstract Algebra: From Numbers to Gröbner Bases*, Cambridge University Press, 2003.

Linear Optimization

Course title	**Linear Optimization**	
Prerequisite courses	Foundations of Matrices and Linear Algebra	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

Optimization is one of the important fields in applied mathematics which leads in interesting connections between Mathematics, Economics, and Industry. In this course, linear optimization in finite-dimensional spaces is addressed.

Course Description

- **Introduction to optimization:** various types of optimization problems and their applications; modeling of optimization problems; geometric method for solving linear programming (LP) problems.
- **Preliminaries from convex analysis and their importance in linear programming:** linear hull, convex hull, affine hull, convex sets; polyhedral sets; hyperplanes and halfspaces; extreme points and extreme directions (definition, characterization, and computation).
- Representation theorems and their consequences; existence of optimal solutions in linear programming.
- **Simplex algorithm:** theoretical, computational, and practical aspects; interpretation of the algorithm from algebraic, geometric, and practical viewpoints; alternative optimal solutions and their computation; degeneracy; convergence properties.
- **Artificial variable techniques:** Big–M method; two-phase method.
- **Optimality and duality:** Karush–Kuhn–Tucker (KKT) theorems for linear programs; linear alternative theorems; duality theory including weak duality, strong duality, and complementary slackness; interpretation and applications of duality; dual simplex algorithm; sensitivity analysis.
- **Transportation and assignment problems:** modeling and solution of transportation problems; simple and composite (transshipment) transportation problems; modeling and solution of the assignment problem.
- **Integer programming:** modeling with integer programming; integer programming solution methods including Branch-and-Bound and Cutting Plane methods.
- **Software and modeling tools:** introduction to software packages and modeling languages for optimization problems such as GAMS, CPLEX, LINGO, and related tools.

References

- M. S. Bazaraa, J. J. Jarvis, and H. D. Sherali, *Linear Programming and Network Flows*, Wiley, 2006.
- K. G. Murty, *Linear Programming*, Wiley, 1983.
- D. Bertsimas and J. N. Tsitsiklis, *Introduction to Linear Optimization*, Athena Scientific, 1997.

Statistical Methods

Course title	**Statistical Methods**	
Prerequisite courses	Probabilty I	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

In this course, a summary of statistical methods used in statistical inference is taught to the student in a simple and introductory way. In addition to statistics students, this course is recommended for the students of computer science and mathematics who seeking a single introductory course in both probability and statistics

Course Description

- Random sample, sample mean distribution and central limit theorem, sample distributions, statistical inference.
- **Estimation methods:** point estimation, interval estimation, confidence interval with large sample size.
- **Introduction to the concepts of hypothesis test:** simple hypothesis test, one-sided hypothesis test, two-sided hypothesis test and maximum likelihood method, hypothesis testing for variance of Normal distribution, hypothesis testing for mean and proportion with larger sample, inference about two means, inference about paired observations, hypothesis testing for two means and proportions in the large population, one-sided analysis of variance, linear regression and correlation coefficient, goodness of fit.
- **Contingency tables:** independence and homogeneity, simple non-parametric test.

References

- J. Behboodian, *An Introduction to Probability and Statistics*, Astanghodsrazavi Publications, 2004.
- A. Parsian, *An Introduction to Probability and Statistics for Science and Engineering*, Isfahan University of Technology Publications, 2009.
- R. Hogg and E. Tanis, *Probability and Statistical Inference*, translated by N. Izaddoostar and H. Pezeshk, Tehran University Publications, 2004.
- R. L. Ott and M. T. Longnecker, *An Introduction to Statistical Methods & Data Analysis*, 7th ed., ISBN 9781305269477, 2015.

Coding Theory

Course title	**Coding Theory**	
Prerequisite courses	Foundations of Combinatorics - Foundations of Matrices and Linear Algebra	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

Familiarity with the introductory and fundamental concepts of Coding Theory.

Course Description

- **Foundations:** review of algebraic concepts including rings, fields, quotient rings, principal ideal domains (PID), polynomial rings, and the division algorithm; vector spaces over finite fields; basis and dimension of a vector space.
- **Introduction to Coding Theory:** basic definitions including alphabet, code, and codeword; Hamming distance and its properties; nearest neighbor decoding; definition of a symmetric channel; criteria based on Hamming distance for error detection and error correction using the nearest neighbor method; statement of the main problem of coding theory and its solution in specific cases; equivalent codes and illustrative examples.
- **Binary Codes and Geometry:** weight of a binary codeword and its relation to Hamming distance; neighborhood and covering radius of a word and of a code; perfect codes, including definitions and examples.
- **Finite Fields:** introduction to finite fields; characteristic of a field (prime characteristic for finite fields and characteristic zero for infinite fields); order of a finite field; field of fractions; irreducible polynomials; construction of finite fields.
- **Linear Codes:** introduction to linear codes; weight of codewords in a linear code and the relationship between minimum weight and minimum distance; generator matrix of a linear code; equivalent linear codes and examples; encoding and decoding using linear codes; Slepian's standard array decoding method; probability of error detection and correction in linear codes; capacity of a binary symmetric channel.
- **Dual Codes and Hamming Codes:** dual codes; parity-check matrix and its role in decoding; standard form of a parity-check matrix; binary Hamming codes, including parameter analysis and decoding methods; extended Hamming codes.
- **Cyclic Codes:** definition of cyclic codes; polynomial representation of linear codes; criteria for identifying cyclic codes using ideals; generator polynomial and generator matrix of a cyclic code; generator polynomial and parity-check matrix of the dual cyclic code; encoding and decoding using generator matrices of cyclic codes and parity-check matrices of their duals.

References

- S. Ling and C. Xing, *Introduction to Coding Theory* (Persian translation), translated by Seyyed Hessamoddin Sharifi and Kobra Ali-Mohammadi, Shahed University Press, 1394 [2015].
- R. Hill, *A First Course in Coding Theory*, Oxford Applied Mathematics and Computing Science

Series, Clarendon Press, 11th ed., 2001.

- S. Roman, *Coding and Information Theory*, Springer, 1992.
- S. Ling and C. Xing, *Coding Theory: A First Course*, Cambridge University Press, 2014.

دانشگاه صنعتی خواجه نصیرالدین طوسی

6th Semester 2020-2021 session (Winter)

Foundation of Geometry

Course title	**Foundation of Geometry**	
Prerequisite courses	Foundations of Mathematics	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Optional Course	

General Purpose

Geometry has always been an inspiration for mathematicians as well as a helpful tool to promote mathematics. It is crucial for science, especially mathematics students, to be systematically familiar with the fundamentals of geometry and know the difference between different aspects of distinct geometries. It would help students in their path from elementary geometry to more advanced geometrical concepts. This course will be helpful to all science, especially Physics and Mathematics students

Course Description

- **Axiomatic system in Geometry:** incidence axioms, betweenness axiom, line segment, ray, Pasch theorem
- **definitions of geometric concepts:** angle, inside, and outside of angle, crossbar theorem, definition of triangle, concept of convexity, axioms of congruence and their elementary corollaries on defining right angle, right triangle congruence theorem, supplementary angles
- **parallelism and perpendicularity:** existence and uniqueness of perpendicular to a line, triangle inequality, midpoint of a segment, angle bisector, perpendicular bisector
- **Dedekind cut:** Dedekind cut of the line, Dedekind axiom of continuity
- **The defect of a triangle:** definition of circle and preliminary properties, intersection of a line and a circletheorem, two circles theorem and Archimedean property, length of segment, measure of angles, defect of a triangle and Saccheri-Legendre theorem, quadrilaterals, rectangles, defect of a triangle and existence of rectangle, Saccheri and Lambert quadrilateral (theorems of this section without proof)
- **Euclidean and non-Euclidean geometry:** Euclidean parallel postulate and its equivalences, Euclidean geometry, sum of angles of a triangle in Euclidean geometry, similar triangles, negation of parallel postulate and hyperbolic geometry, angle of parallelism, BolyaiLobachevsky theorem, area in hyperbolic geometry, Beltrami Klein model for hyperbolic geometry, Poincaré disk model, categoricity of Euclidean and hyperbolic geometry

References

- M. Greenberg, *Euclidean and Non-Euclidean Geometries*, W. H. Freeman and Company, 1993.
- E. Moise, *Elementary Geometry from an Advanced Standpoint*, Addison–Wesley Publishing Company, 1990.
- R. Hartshorne, *Geometry: Euclid and Beyond*, Springer-Verlag, 2000.

Graph Theory and Its Applications

Course title	**Graph Theory and Its Applications**	
Prerequisite courses	Foundations of Combinatorics	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Optional Course	

General Purpose

The field of graph theory has been grown extensively in the 20th century. One of the main reasons of such a huge growth is the ability of graphs in modeling different types of problems as well as connections of graphs to computer science. In this course the student get familiar with basic concepts of graphs theory as well as some of its connections with other fields.

Course Description

- **Basic concepts:** Graphs and their representations (incidence and adjacency matrices, adjacency lists), subgraphs, paths and connectivity, isomorophism of graphs, operations on graphs, digraphs and tournaments.
- **Connectivity**: cut-vertices and cut-edges, connectivity and edgeconnectivity, blocks, cutsets and edge-cutsets.
- **Trees:** directed and rooted trees, traversing trees, depth-search and breadth-search, some enumeration problems in graphs, finding minimum spanning trees in a connected weighted graph (Kruskal and Prim Algorithms).
- **Distances of vertices:** application of BFS in finding distance of vertices, finding shortest paths in weighted graphs (Dijestra algorithm).
- **Matchings and independent sets:** independent sets of vertices, independent sets of edges, Berge theorem about maximum matchings, matchings and factors, Hall theorem, maximum matchings in bipartite graphs, Petersen theorem, Tutte-Berge theorem.
- Eulerian and Hamiltonian graphs, Euler theorem, Dirac theorem, Ore theorem, Chinese postman problem, traveling salesman problem.
- **Network flows:** max-flow min-cut theorem, Menger theorem.
- **Colorings of graphs:** vertex-coloring, chromatic number, Brooks theorem,edge-coloring, edgechromatic numbers, Vizing theorem, chromatic polynomial.
- **Planarity:** planar graphs, Euler formula and some consequences, dual of a plane graph, Kuratowski theorem.

References

- J. A. Bondy and U. S. R. Murty, *Graph Theory*, Springer, 2008.
- R. Diestel, *Graph Theory*, Springer, 2006.
- D. B. West, *Introduction to Graph Theory*, Prentice Hall, 2001.
- R. Balakrishnan and K. Ranganathan, *A Textbook of Graph Theory*, Springer, 2000.

Stochastic Processes

Course title	**Stochastic Processes**	
Prerequisite courses	Probability I	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

In this course, the basic and theoretical principles of stochastic processes are taught.

Course Description

- **Markov chains:** transition function; one-step and multi-step transition matrices; Chapman–Kolmogorov equation; initial distribution; absolute and joint distributions; hitting times and mean first passage times; transient and recurrent states (positive and null recurrent); absorption probabilities; birth and death chains; branching chains and queuing; state space decomposition; random walk and Gambler's Ruin problem.
- **Stationary distributions:** definition and properties of stationary distributions; irreducible chains; birth and death chains; expected number of visits to recurrent states; mean return time; methods for computing stationary distributions; Monte Carlo methods; estimation of the transition matrix; estimation of stationary distributions.
- **Markov pure jump processes:** properties of pure jump processes; applications of pure jump processes in birth and death processes and queuing systems.
- **Poisson processes:** definition and properties of the Poisson process; homogeneous and non-homogeneous Poisson processes; decomposition of the Poisson process; arrival times and inter-arrival times; relationship with exponential, gamma, geometric, uniform, and beta distributions; conditional distribution of inter-arrival times given the number of events and relation to order statistics; compound Poisson process; applications in renewal theory and risk theory; restarted Poisson process.
- **Continuous-time Markov processes:** general properties; birth and death processes; Brownian motion and basic properties.
- **Basic concepts of stochastic processes:** finite-dimensional distributions; stationarity; homogeneous and independent samples; Markov property; Bernoulli and binomial processes and their properties.
- **Generating functions:** review of generating functions; sum of a random number of random variables.
- **Computational and simulation methods:** simulation of Bernoulli counting processes; simulation of random walks; simulation of Poisson processes; simulation of Markov chains.
- **Introduction to Markov Chain Monte Carlo (MCMC) methods using software.**

References

- P. Brémaud, *Markov Chains, Gibbs Fields, Monte Carlo Simulation and Queues*, Springer, New York, 1999.

- N. Bhat and K. Miller, *Elements of Applied Stochastic Processes*, 3rd ed., John Wiley & Sons, 2002.
- P. G. Hoel, S. C. Port, and C. J. Stone, *Introduction to Stochastic Processes*, Houghton Mifflin Company, 1972.
- S. Karlin and H. M. Taylor, *An Introduction to Stochastic Modeling*, Academic Press, 1994.
- S. M. Ross, *Introduction to Probability Models*, Academic Press, 2014.

دانشگاه صنعتی خواجه نصیرالدین طوسی

Theoretical Foundations of Computability

Course title	**Theoretical Foundations of Computability**	
Prerequisite courses	Foundations of Mathematics	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

Discussion on core concepts: "Problem", "Calculation", "Solution Method", "Computational Model", "Algorithm", and explaining the main objectives of this branch. Approaching calculation theory from the perspective of computational models and presenting the concept of automata as a computational model with fixed memory, proposing types of automata (DFA, NFA, ANFA) and their relevant spaces, introducing regular expressions, defining regular languages and the equivalence of these definitions, definition of minimal automata and the method to obtain it (without proof). Non-Regular Languages (Presenting necessary and sufficient conditions, pumping lemma), presenting the Pushdown Automata model, presenting the concept of Grammar, Context-Free Grammar and the equivalence of these concepts, defining Chomsky Normal Form, Context-Free pumping spaces for languages independent of context, familiarity with the Turing Machine model, stating various types of this model and the equivalence of this concept with arbitrary Grammars.

Course Description

- **Regular Languages & Automata:**
  - Deterministic Finite Automata (DFA) and Non-deterministic Finite Automata (NFA).
  - NFA with λ-moves (silent transitions) and their equivalence.
  - Regular Expressions and their relationship to Finite Automata.
  - Simulation concepts and closure properties.
  - Pumping Lemma for Regular Languages and non-regular languages.
  - Minimal Automata.
- **Context-Free Languages:**
  - Context-Free Grammars (CFG) and derivation trees.
  - Ambiguity in grammars.
  - Pushdown Automata (PDA) and their types (Deterministic vs. Non-deterministic).
  - Equivalence of CFG and PDA.
  - Chomsky Normal Form.
  - Pumping Lemma for Context-Free Languages.
- **Turing Machines & Computability:**
  - Turing Machines and Recursive & Recursively Enumerable languages.
  - Variations of Turing Machines and their equivalence.
  - Universal Turing Machine.
  - Unrestricted Grammars and Context-Sensitive Grammars (Linear Bounded Automata).
  - Decidability, Undecidability, and the Post Correspondence Problem (PCP).

References

- R. Greenlaw and H. J. Hoover, *Fundamentals of the Theory of Computation: Principles and Practice*, Morgan Kaufmann, 1998.
- J. Martin, *Introduction to Languages and the Theory of Computation*, McGraw-Hill, 2010.
- M. Sipser, *Introduction to the Theory of Computation*, Thomson Course Technology, 2006.

دانشگاه صنعتی خواجه نصیرالدین طوسی

Elementary Cryptography

Course title	**Elementary Cryptography**	
Prerequisite courses		**Course Nature**
		Theoretical ■ Practical □
		Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

Course Description

- **Security Fundamentals:** Analysis of the OSI security architecture, the CIA triad (Confidentiality, Integrity, Availability), and the fundamental models for network security and design principles.
- **Classical Encryption:** Study of symmetric cipher models and historical techniques, including substitution (Caesar, Playfair, Hill, Vigenère) and transposition ciphers.
- **Number Theory Foundation:** Comprehensive study of divisibility, the Euclidean Algorithm, modular arithmetic, and prime numbers (Fermat's and Euler's Theorems).
- **Symmetric Block Ciphers:** Technical examination of the Feistel Cipher structure, the Data Encryption Standard (DES), and the Advanced Encryption Standard (AES) transformation functions.
- **Block Cipher Operation:** Analysis of Multiple Encryption (Triple DES) and modes of operation such as ECB, CBC, CFB, OFB, and CTR for various application environments.
- **Public-Key Cryptography Principles:** Introduction to asymmetric cryptography, the concept of trapdoor one-way functions, and the comparison between symmetric and public-key systems.
- **The RSA Algorithm:** Detailed study of the RSA scheme, including key generation, encryption/decryption processes, and its security against various mathematical attacks.
- **Key Exchange and Discrete Logs:** Mechanics of the Diffie-Hellman Key Exchange and its security foundation in the Discrete Logarithm Problem.
- **Elliptic Curve Cryptography (ECC):** Introduction to the mathematics of elliptic curves over finite fields and their application in modern, high-security, low-overhead cryptographic systems.
- **Cryptographic Hash Functions:** Applications of hash functions for message integrity, requirements for collision resistance, and the study of SHA-512 and SHA-3.
- **Message Authentication:** Analysis of Message Authentication Codes (MACs) and the construction of HMAC to ensure data origin authentication and integrity.

References

- W. Stallings, *Cryptography and Network Security: Principles and Practice*, 7th Edition, Pearson, 2017.
- D. R. Stinson, *Cryptography: Theory and Practice*, 3rd Edition, Chapman & Hall / CRC, 2006.
- W. Trappe and L. C. Washington, *Introduction to Cryptography with Coding Theory*, 3rd Edition, Pearson, 2020.
- J. Katz and Y. Lindell, *Introduction to Modern Cryptography*, 3rd Edition, CRC Press, 2020.
- J. Hoffstein, J. Pipher, and J. H. Silverman, *An Introduction to Mathematical Cryptography*,

Springer, 2008.

- T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd Edition, Wiley-Interscience, 2006.
- H. Beker and F. Piper, *Cipher Systems: The Protection of Communications*, International Thomson Publishing Ltd., London, 1982.

دانشگاه صنعتی خواجه نصیرالدین طوسی

Specialized English for Mathematics

Course title	**Specialized English for Mathematics**	
Prerequisite courses		**Course Nature**
		Theoretical ☐ Practical ☐ Computer Science Minor ☐
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

Familiarity with English texts and standard mathematical terminology in the main and foundational undergraduate topics of mathematics.

Course Description

- **Foundations of Mathematics:** reading English texts related to elementary logic; propositions and quantifiers; elementary set theory; basic operations on sets.
- **Mathematical Analysis:** reading English texts related to the real number system; metric spaces and related concepts; sequences; functions; continuity.
- **Linear Algebra:** reading English texts related to vector spaces; bases; dimension; linear transformations.
- **Algebra:** reading English texts related to groups; order of elements; cyclic groups; permutation and alternating groups; polyhedral groups; subgroups.
- **Operations Research:** reading English texts related to the history of operations research; familiarity with introductory optimization problems.

References

- J. E. Hutchinson, *Introduction to Mathematical Analysis*, ANU, 2000.
- S. Waner, *Introduction to Group Theory*, Lecture Notes, 2003.
- J. Heffron, *Linear Algebra*, Lecture Notes, 2012.
- P. R. Murthy, *Operations Research*, New Age International Publishers, 2007.

دانشگاه صنعتی خواجه نصیرالدین طوسی

7th Semester 2021-2022 session (Fall)

Algebra

Course title	**Algebra**	
Prerequisite courses	Foundations of Algebra	**Course Nature**
		Theoretical ■ Practical □
		Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

- Group action on sets and using it to prove some of the Sylow theorems.
- Series of groups and solvable groups.
- Factorization theory and polynomial rings (PID, UFD, ED).
- Elementary field extensions, ruler and compass constructions, impossibility of squaring the circle, doubling the cube, and trisecting an angle.
- Statement of Galois' theorems (without proof) accompanied by several examples.

Course Description

Groups

- Review of material on Group Theory from the "Foundations of Algebra" course .
- Types of group series (subnormal, normal, composition, etc.) and examples of them .
- Proof of the Schreier and Jordan-Hölder theorems, accompanied by examples .
- Solvable groups and nilpotent groups, subgroups, quotient groups, commutator subgroup, etc. .
- Group action on sets and its properties .
- p-groups, proof of some Sylow theorems, applications of Sylow theorems .

Factorization Theory and Polynomial Rings

- Review of material on Polynomial Rings from the "Foundations of Algebra" course .
- Irreducible polynomials and irreducible elements .
- Principal Ideal Domains (PID), examining $F[x]$ with the idea of principal ideals .
- Unique Factorization Domains (UFD), examining $F[x]$.
- Euclidean Domains (ED), examining $F[x]$.

Field Extensions

- Review of material from Linear Algebra regarding finite and finitely generated extensions (field extensions) .
- Algebraic extensions, minimal polynomials .
- Algebraically closed fields .

Famous Impossibilities

- Ruler and compass constructions, constructible numbers and fields .
- Proving the impossibility of squaring the circle, doubling the cube, and trisecting an angle .

Galois Theory

- Preliminaries required to state Galois' theorems .
- Statement of Galois' theorems (without proof) accompanied by examples .

References

- D. S. Dummit and R. M. Foote, *Abstract Algebra*, Wiley, 2004.
- J. B. Fraleigh, *A First Course in Abstract Algebra*, 6th Edition, Addison-Wesley, 2002.
- T. W. Hungerford, *Abstract Algebra: An Introduction*, Saunders College Pub., 1996.
- I. N. Herstein, *Abstract Algebra*, Macmillan Company, 1989.
- I. N. Stewart, *Galois Theory*, CRC Press, 2015.

دانشگاه صنعتی خواجه نصیرالدین طوسی

Partial Differential Equations

Course title	**Partial Differential Equations**	
Prerequisite courses	Calculus II - Differential Equations	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

To familiarize students with Fourier analysis and the classification and solution methods for second-order linear partial differential equations, including elliptic, parabolic, and hyperbolic equations.

Course Description

- **Introduction and First-Order Equations:** Initial definitions and notation; origins of PDEs in physical problems; modeling; Methods of characteristics and Lagrange for solving first-order PDEs; Cauchy's method of characteristics; Charpit's method and Jacobi's method for solving nonlinear first-order equations.
- **Second-Order Equations and Classification:** Origin and classification of second-order PDEs with constant and variable coefficients; normal forms; converting Hyperbolic, Parabolic, and Elliptic equations to canonical forms and solving them.
- **Fourier Analysis and Wave Equation (Hyperbolic):** Review of Fourier series, Fourier integrals, and Fourier transforms; The Wave Equation: D'Alembert's solution; separation of variables; wave equation on an infinite line; Conservation laws in one dimension.
- **Potential Theory (Elliptic Equations):** Derivation of Laplace and Poisson equations; fundamental solutions; Green's functions; Boundary value problems (Dirichlet and Neumann) for rectangles, circles (interior and exterior problems), and spheres; Laplace equation in cylindrical and spherical coordinates.
- **Heat Equation (Parabolic Equations):** Modeling heat conduction; separation of variables for initial-boundary value problems; Maximum principle and uniqueness of solutions; Heat equation on infinite and semi-infinite lines.
- **Integral Transform Applications:** Laplace transform methods for PDEs; Fourier transform methods (Exponential, Sine, and Cosine transforms); Applications to heat conduction in solids, heat flow in infinite rectangles, mechanical vibrations, transverse vibrations of elastic beams, and potential theory in half-planes and infinite strips.

References

- M. Hesaraki and M. Fotouhi, *Partial Differential Equations*, Sharif University of Technology, 2010.
- Z. Afsharnezhad, *Partial Differential Equations and Elementary Partial Derivatives*, Parvin Publications, 1994.
- I. N. Sneddon, *Elements of Partial Differential Equations*, McGraw-Hill, 1985.
- R. Haberman, *Elements of Applied Partial Differential Equations*, Prentice Hall, 1983.
- J. N. Sharma and K. Singh, *Partial Differential Equations for Engineers and Scientists*, Narosa Publishing House, 2000.

Numerical Linear Algebra

Course title	**Numerical Linear Algebra**	
Prerequisite courses	Foundations of Matrices and Linear Algebra	**Course Nature** Theoretical ■ Practical □ Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

Linear systems of equations appear in many problems such as image and signal processing, control and statistic processes. In this course, numerical algorithms to solve linear systems of equations, finding eigenvalues and eigenvectors, solving minimum least squares problems and other matrix problems are studied.

Course Description

- **Preliminaries of Linear Algebra:** Review of vector spaces, linear independence, basis; inner products; vector and matrix norms; range and null spaces; rank; similarity; special matrices.
- **Direct Methods for Linear Systems:** Floating point arithmetic and error analysis; Gaussian elimination with pivoting (partial and complete); Matrix decomposition methods: LU decomposition and Cholesky (LL^T) decomposition for positive definite matrices; Computational cost (operation counts) and error analysis.
- **Sensitivity and Stability:** Numerical stability of algorithms; well-posed and ill-posed problems; Condition numbers and sensitivity analysis of linear systems; Stability of LU decomposition.
- **Least Squares and Orthogonalization:** Matrix geometric interpretation; QR decomposition (using Householder transformations and Givens rotations); Solving the linear least squares problem and data fitting; Singular Value Decomposition (SVD) basics (optional/if time permits).
- **Iterative Methods for Linear Systems:** Basic iterative methods: Jacobi, Gauss-Seidel, and relaxation methods; Convergence and stability analysis of iterative methods.
- **Eigenvalues and Eigenvectors:** Numerical methods for computing eigenvalues and eigenvectors; Power method and Inverse Power method; Rayleigh quotient iteration; QR algorithm for eigenvalues; Reduction to Hessenberg or tridiagonal forms (Householder method).

References

- B. N. Datta, *Numerical Linear Algebra and Applications*, SIAM, 2nd edition, 2010.
- L. N. Trefethen and D. Bau, *Numerical Linear Algebra*, SIAM, 3rd edition, 1997.
- A. Stuart and J. Voss, *Matrix Analysis and Algorithms*, Lecture notes, 2009.

Logic Circuits

Course title	**Logic Circuits**	
Prerequisite courses		**Course Nature**
		Theoretical ■ Practical □
		Computer Science Minor ■
Number of credits	3	
Number of hours	51	
Course classification	Major Course	

General Purpose

This course provides the design skills and theoretical knowledge required for the design, simulation, and construction of basic combinational and sequential logic circuits. Upon successful completion of this course, the student will be able to use the tools presented to design and implement a set of combinational and sequential circuits, such as a simple calculator or a digital counter.

Course Description

- **Number Systems and Codes:** Digital systems; binary numbers; number-base conversions; octal and hexadecimal numbers; complements of numbers; signed binary numbers; binary codes (BCD, Gray code, ASCII); binary storage and registers; binary logic.
- **Boolean Algebra and Logic Gates:** Basic definitions; axioms and theorems of Boolean algebra; Boolean functions; canonical and standard forms (minterms and maxterms); logic operations; digital logic gates; integrated circuits.
- **Gate-Level Minimization:** The map method (Karnaugh map); three- and four-variable maps; product-of-sums simplification; don't-care conditions; NAND and NOR implementation; Exclusive-OR function.
- **Combinational Logic:** Combinational circuits analysis and design procedures; binary adder-subtractor; decimal adder; binary multiplier; magnitude comparators; decoders and encoders; multiplexers and demultiplexers.
- **Synchronous Sequential Logic:** Sequential circuits; latches (SR, D); flip-flops (Edge-Triggered, JK, T); analysis of clocked sequential circuits; state equations, state tables, and state diagrams; design procedure for sequential circuits.
- **Registers and Counters:** Registers; shift registers (serial/parallel transfer); ripple counters; synchronous counters; ring counters; Johnson counters.
- **Memory and Programmable Logic:** Random-Access Memory (RAM); Read-Only Memory (ROM); Programmable Logic Array (PLA); Programmable Array Logic (PAL).
- **Algorithmic State Machines (ASM):** ASM chart; timing considerations; control implementation; design with multiplexers.

References

- M. Morris Mano and Michael D. Ciletti, *Digital Design, with an introduction to the Verilog HDL*, Pearson, 2013 (5th Edition)

8th Semester 2021-2022 session (Winter)

Elementary Algebraic Geometry

Course title	**Elementary Algebraic Geometry**	
Prerequisite courses	Foundations of Algebra - Foundations of Matrices and Linear Algebra	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Optional Course	

General Purpose

Introductory familiarity with the main structures of algebraic geometry, namely affine varieties, projective varieties, and quasi-projective varieties, and their algebraic properties.

Course Description

- **Algebraic Foundations:** Elementary properties of multivariate polynomial rings and their ideals; Hilbert's Basis Theorem.
- **Affine Varieties:** Affine space; affine varieties and Zariski topology; Hilbert's Nullstellensatz (without complete proof); correspondence between varieties and ideals; decomposition of varieties into irreducible components.
- **Functions on Affine Varieties:** Polynomial functions on affine varieties; coordinate ring; relationship between isomorphism of affine varieties and isomorphism of coordinate rings; function field of an irreducible affine variety.
- **Projective Varieties:** Projective space; homogeneous ideals; projective varieties; homogeneous coordinate ring; Hilbert's Nullstellensatz for projective varieties; affine cone over a projective variety; homogenization of an ideal and projective closure of an affine variety.
- **Maps and Quasi-Projective Varieties:** Maps between varieties; definition of quasi-projective varieties; affine base for the Zariski topology on a quasi-projective variety.
- **Regular Functions and Sheaves:** Regular functions on a quasi-projective variety; ring of regular functions; sheaf of regular functions on a quasi-projective variety (without proof); stalk of the structure sheaf at a point; function field of a quasi-projective variety.
- **Products and Local Properties:** Product of two affine varieties and its coordinate ring; Segre embedding and product of two projective (and quasi-projective) varieties; Dimension of a variety; Dimension Theorem (without proof); Tangent space at a point on an affine variety; definition of a smooth (non-singular) point; theorem on the set of singular points (without proof).
- **Computational Methods:** Computational methods in algebraic geometry (Gröbner bases and their applications).

References

- G. Harder, *Lectures on Algebraic Geometry II: Basic Concepts, Coherent Cohomology, Curves and their Jacobians*, Vieweg+Teubner, 2011.
- D. P. Patil and U. Storch, *Introduction to Algebraic Geometry and Commutative Algebra*, IISc Lecture Notes, World Scientific, 2010.

- D. Perrin, *Algebraic Geometry: An Introduction*, Springer-Verlag, 2008.
- K. Kendig, *Elementary Algebraic Geometry*, Springer-Verlag, 1977.
- B. Hassett, *Introduction to Algebraic Geometry*, Cambridge University Press, 2007.
- M. Reid, *Undergraduate Algebraic Geometry*, Cambridge University Press, 1988.

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Complex Functions

Course title	**Complex Functions**	
Prerequisite courses	Foundations of Mathematical Analysis	**Course Nature** Theoretical ■ Practical ☐ Computer Science Minor ☐
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

The theory of functions of a complex variable is an important part of pure mathematics, engineering mathematics, and mathematical physics. It is helpful in many branches of mathematics, including algebraic geometry, number theory, analytic combinatorics, applied mathematics, as well as in physics, including the branches of hydrodynamics, thermodynamics, and particularly quantum mechanics.

Course Description

- **Complex Numbers System:** The Complex Number System; fundamental operations; conjugate; absolute value; polar form; De Moivre's theorem; roots of complex numbers; Euler's formula; polynomial equations; the n-th roots of unity; point set topology; continuity; theorems on limits; uniform convergence of functions.
- **Analytic Functions:** Derivatives; analytic functions; Cauchy–Riemann equations; harmonic functions; geometric interpretation of the derivative; rules for differentiation; derivatives of elementary functions; higher order derivatives.
- **Complex Integration:** Curves; complex line integrals; properties of integrals; change of variables; simply and multiply connected regions; Jordan curve theorem; Green's theorem in the plane; complex form of Green's theorem; Cauchy's theorem; the Cauchy–Goursat theorem; Morera's theorem; indefinite integrals; integrals of special functions; some consequences of Cauchy's theorem; Cauchy's integral formulas; Cauchy's inequality; Liouville's theorem; Fundamental Theorem of Algebra; Gauss' mean value theorem; maximum modulus theorem; minimum modulus theorem; the argument theorem; Rouché's theorem.
- **Infinite Series:** Sequences of functions; series of functions; uniform convergence of sequences and series; power series; Taylor's theorem; some special series; Laurent's theorem; classification of singularities; analytic continuation.
- **The Residue Theorem & Evaluation of Integrals and Series:** Residues; calculation of residues; the Residue theorem; evaluation of definite integrals; special theorems used in evaluating integrals; the Cauchy principal value of integrals; differentiation under the integral sign (Leibniz's Rule); summation of series; Mittag–Leffler's expansion theorem; some special expansions.
- **Conformal Mapping:** Transformations or mappings; Jacobian of a transformation; complex mapping functions; conformal mapping; some general transformations; the linear transformation; the bilinear or fractional transformation; mapping of a half plane onto a circle; the Schwarz–Christoffel transformation.

References

- M. Hesaraki and M. R. Pournaki, *Functions of Complex Variables* (in Persian), Fatemi, Iran, 2010.

- J. B. Conway, *Functions of One Complex Variable I*, Springer, 1995.
- M. Spiegel, S. Lipschutz, J. Schiller, and D. Spellman, *Schaum's Outline of Complex Variables*, 2nd ed., McGraw-Hill, 2009.

دانشگاه صنعتی خواجه نصیرالدین طوسی

General Topology

Course title	General Topology	
Prerequisite courses	Foundations of Mathematical Analysis	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Core Course	

General Purpose

Introductory familiarity with the fundamental concepts of general topology, including the construction of topological spaces, continuity, compactness, connectedness, and separation axioms, as well as an introduction to metric spaces, Euclidean topology, and topological groups.

Course Description

- **Introduction and Basic Definitions:** Definition of various topologies using open set axioms (or equivalently, closed sets); closure and Kuratowski closure operator; examples of different spaces (finite, discrete, indiscrete, metric, order). Neighborhood of a point, interior point, limit point, boundary point, adherent point; closure of a set and dense subsets. Basis and subbasis; local basis (basis at a point); first and second countable spaces; separable spaces. Separation axioms: T_0, T_1, and T_2 (Hausdorff) spaces.
- **Continuous Functions and Constructing New Spaces:** Continuity of a function at a point; continuous functions; open maps; closed maps; homeomorphisms. Subspaces; product spaces and sum (disjoint union) spaces; embeddings; identification (quotient) maps and quotient spaces; strong and weak topologies.
- **Connectedness and Compactness:** Compactness; compact spaces with order topology. Path connectedness and path components; local connectedness and local path components.
- **Separation Axioms:** Regular spaces; Normal spaces; Urysohn's Lemma.
- **Filters and Nets:** Filters; Nets; Alexander Subbase Theorem; proof of Tychonoff's Theorem.
- **Metric Space Topology:** Metrizability theorems; Baire Category Theorem; Space-filling curves.
- **Euclidean Space Topology:** Brouwer Fixed Point Theorem; Invariance of Domain; Jordan Curve Theorem; Graphs and Kuratowski's Theorem.
- **Topological Groups (Time Permitting):** Topological groups, subgroups, and quotient groups; action of a topological group on a topological space and homogeneous spaces. Birkhoff-Kakutani theorem on metrizability of topological groups; matrix topological groups.

References

- Bourbaki, Nicolas. *Elements of Mathematics. General Topology. Part 1.* Hermann, Paris; Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont., 1966.
- Bourbaki, Nicolas. *Elements of Mathematics. General Topology. Part 2.* Hermann, Paris; Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont., 1966.
- Hu, Sze-tsen. *Elements of General Topology.* Holden-Day, Inc., San Francisco, Calif.-London-Amsterdam, 1964.
- Willard, Stephen. *General Topology.* Dover Publications, Inc., Mineola, NY, 2004. (Reprint of the

1970 original, Addison-Wesley).

- Munkres, James R. *Topology: A First Course*. Prentice-Hall, Inc., Englewood Cliffs, N.J., 1975.

دانشگاه صنعتی خواجه نصیرالدین طوسی

Mathematical Software

Course title	**Mathematical Software**	
Prerequisite courses	Foundations of Numerical Analysis	**Course Nature** Theoretical ■ Practical □ Computer Science Minor □
Number of credits	3	
Number of hours	51	
Course classification	Optional Course	

General Purpose

Familiarity with one or more mathematical software packages (such as MATLAB, MAPLE, MATHEMATICA, etc.) for performing calculations, programming, and analyzing computational errors in mathematical software.

Course Description

- **Software Environment and Basics:** Familiarity with the environment of one or several mathematical software packages; data types and error analysis within the software.
- **Programming and Visualization:** Programming within the software; plotting 2D and 3D graphs.
- **Numerical and Symbolic Computations:** Matrix operations; solving systems of linear equations; polynomials; interpolation; numerical integration; differential equations; Fourier transform and Laplace transform.
- **Advanced Topics:** Selected new mathematical topics (time permitting).
- **Mathematical Typesetting:** Familiarity with software for mathematical text editing, such as LaTeX.

References

- Mohammadi, Alireza. *Comprehensive Guide to MATLAB and Simulink.*
- Ziaee, Majid (Trans. & Comp.). *Complete Reference to MAPLE.*
- Heidari, Abdolrahman. *Image Processing in MATLAB.*
- McAndrew, Alasdair. *Introduction to Digital Image Processing with MATLAB.* (Translated by Arash Rahmani Zadeh and Mousa Salimi).
- Chapra, Steven C. *Applied Numerical Methods with MATLAB.*